Workflow, Modelling & Webscraping: BEE2041

Data Science for Economics

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Project Management & Workflow 

Today 📋

1. Project management and replicability
2. Workflow and project structure
3. Automation with `make`
4. Documentation

Some Background on Project Management 🏗️

- The nature of our projects will likely hugely vary, but it is not uncommon for projects to last **multiple years** and involve **multiple people**
- The needs of projects written for ourselves and those done in teams differ
- Given long time frames, it is worth investing in efficient work and communication:
  - Good **workflow**
  - Good **project structure** and code style
  - **Replicability**
  - Good **documentation**

Project Management: Why it Matters

In this block we will talk through these issues and connect them to an application in causal ML and webscraping

- These will be stylised examples
- But they give us reason to pause and examine each part of the toolchain carefully
- Optimised to data analysis in economics: in line with our particular interests in this course
- *But* most principles generalise far beyond economics

Useful Resources 📚

Two particularly good references:

1. "Coding for Economists" (Turrell, 2023): freely available, updated, Python-centric
2. "Code and Data for the Social Sciences: A Practitioner's Guide" (Gentzkow and Shapiro, 2014) – a classic, research-focused but broadly applicable

Both are worth bookmarking; we will draw on them throughout these classes.

Workflow and Project Structure 

What is Workflow?

The programming languages we use are embedded in a more general **workflow** – which may include version control, typesetting/markdown, replication, and more

- Building a standardised workflow will be useful across **all** your projects
- It will vary depending on the project, collaborators, work locations, etc.
- At a minimum, workflow should be designed to:
  - **(a)** Efficiently permit replication
  - **(b)** Avoid manual functions that potentially introduce errors (e.g., copy & paste)

A Stylised Economic Data Project

A typical economic data analysis project might consist of:

1. **Version control**: tracking all changes
2. **Data collection**: often scraping or API calls
3. **Data cleaning**: wrangling raw data into usable form
4. **Analysis and generation of results**: running models
5. **Presentation**: tables, figures, reports
6. **Replication materials**: packaging everything for others

Tools for Each Step

It is very unlikely that you would want to use a single tool for all steps. A realistic project might look like:

Step	Tool
Version control	**git / GitHub**
Data collection	**Python (webscraping, APIs)**
Data cleaning	**Python + Pandas**
Analysis	**Python + scientific libraries**
Presentation	**Quarto / LaTeX**
Replication	**Makefile or similar**

Why Automate the Full Pipeline?

Ideally we would like to:

1. **Automate** the process completely or as much as possible
    - What if next month you want to update the data at step 1?
    - What if a collaborator suggests a different cleaning step?
2. **Remember** what was done in previous versions to revert if needed

> **Note**
>
> The first is about designing an efficient replication process (e.g., a Makefile); the second is version control (git), which we have already covered.

Workflow: `make` 🛠️

`make` is a Unix tool that specifies the full path a project follows: rather than running steps one by one, you run the Makefile with a **single command**

- The key value-add: `make` only runs **steps that are necessary**
- If nothing prior to step x has been altered, `make` detects this and only runs steps $x, x+1, \ldots$
- A very complete discussion of Make with examples is provided by Jesús Fernández-Villaverde
- Gentzkow & Shapiro also have a nice discussion of shell script alternatives (chapter 2 here)
- Do not feel like you must do this (it is the advanced solution), but you need *something* to oversee the entire project

A Simple Makefile Example

```
# Makefile for a data analysis project

all: output/results.pdf

data/clean.csv: src/clean.py data/raw.csv
    python src/clean.py

output/results.csv: src/analysis.py data/clean.csv
    python src/analysis.py

output/results.pdf: report/paper.qmd output/results.csv
    quarto render report/paper.qmd
```

Running `make` from the project root re-runs **only** what has changed. This is the foundation of a reproducible pipeline.

Modern Alternatives: Snakemake & DVC

`make` is powerful but dated. Modern alternatives exist:

- **Snakemake**: Python-native workflow manager, popular in bioinformatics and increasingly in economics; handles parallelism cleanly
- **DVC (Data Version Control)**: extends git to handle large data files and ML pipelines; integrates with cloud storage
- **Prefect / Airflow**: for production-grade orchestration

For many research projects, we may be happy with just some runner file like `runAll.py`, but we need something to automate “build”.

Documentation 

Documentation: Why It Matters

> **Tip**
>
> “You will probably spend more time **reading** code than **writing** code.” — Jesús Fernández-Villaverde

Research projects should be documented extensively:

- **Globally** — using a `README` file at the project root
- **Locally** — commenting within scripts

Both are important; neither alone is sufficient.

Commenting in Scripts

All serious programming languages allow inline comments:

- Comments are instructions **for human readers** – the machine ignores them (well, it used to…)
- Should be used throughout wherever something is not immediately obvious
- At minimum, **section-level explanations** should always be present
- Comments should be maintained alongside code: stale comments are misleading
- **Err on the side of verbosity** future you will thank present you!

```
# ── Section 1: Load and merge raw data ──────────────────────────────
# We merge the household survey (hhsurvey) with the administrative records (admin)
# using the household ID (hh_id). Some households appear only in one source;
# we keep all observations and flag missings in the merged indicator below.

df = pd.merge(hhsurvey, admin, on='hh_id', how='outer', indicator=True)
```

README Files

A `README` file in the main directory gives **global instructions** for a project:

- Explains procedures followed from start to finish
- Lists external libraries/dependencies needed
- Describes the directory structure
- Provides contact details and citation information

> **Note**
>
> This should be the **first thing you look for** when you download someone else's code. It may be called `README`, `README.txt`, `README.md`, ...

A good short resource on README best practices is available here.

A Good Directory Structure

Consistency in structure matters as much as content. A typical economic project might look like:

```
project/
├── README.md
├── Makefile
├── data/
│   ├── raw/          ← never modified after collection
│   └── clean/        ← output of cleaning scripts
├── src/
│   ├── 01_clean.py
│   ├── 02_analysis.py
│   └── 03_figures.py
├── output/
│   ├── tables/
│   └── figures/
└── report/
    └── paper.qmd
```

The key principle: **raw data is sacred**. Never overwrite it (consider limiting permissions).

Style, Testing & Debugging 

This section📋

1. Consistent programming style
2. Automation and abstraction
3. Testing, debugging, and profiling

Programming Style 🎨

Consistent Style

There are many frequently used stylistic choices in programming – often not enforced by syntax but still very important:

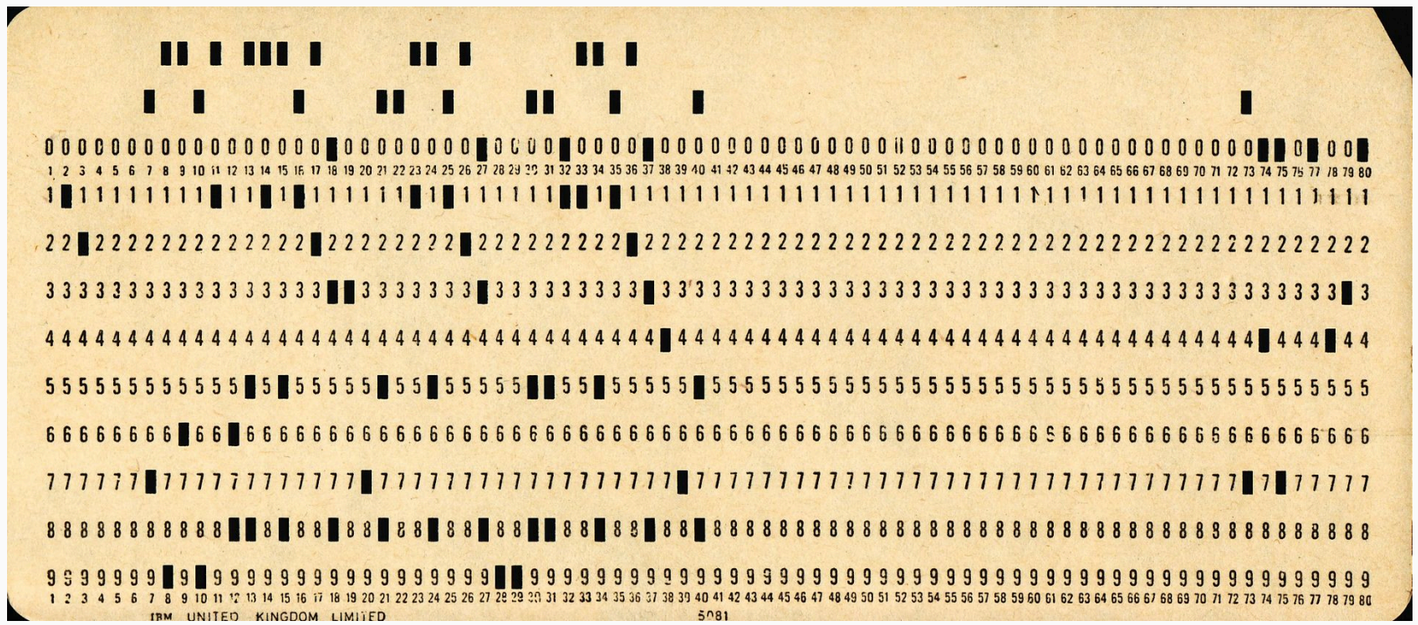
- People develop preferred **directory structures** for projects
- **Delimiter choices** for data (commas, tabs, pipes)
- **Spacing** in code is often user-specific (though some languages enforce rules)
- **Intelligent variable naming** simplifies work considerably

> **Tip**
>
> Even if there are no rules, **internal consistency** is key: it enables re-use of standardised code and reduces cognitive load.

Spacing and Line Length

In some languages, spacing is syntactically significant:

- Python uses **indentation** to delimit loops and conditional blocks
- Fortran reserves initial spaces for particular statements
- The convention of not exceeding **80 characters per line** is very common. We may ask: Why?



Institutions are ingrained and slow to change!

Python Style: PEP 8 & Beyond

Python has an official style guide: PEP 8

Key recommendations: - 4 spaces for indentation (never tabs) - Maximum line length of 79 characters (PEP 8) or 88 (Black formatter) - Two blank lines between top-level functions/classes - One blank line between methods

Tooling that enforces this automatically:

- `black`: opinionated auto-formatter (“the uncompromising code formatter”)
- `ruff`: extremely fast linter + formatter-
- `flake8`: linter (catches style and logic errors)

Variable Naming

Variable naming is itself a form of documentation:

- Good naming makes things **dramatically** easier; bad naming is a disaster
- Consider a binary variable for sex at birth — potential names: `{var1, gender, female}`
- Conventions: `snake_case` (Python standard) or `camelCase` (more common in JS/Java)
- Consistency within a project is essential

```
# Bad
x = pd.read_csv('data.csv')
df2 = x[x['a'] > 0]

# Good
raw_survey = pd.read_csv('data/raw/household_survey_2023.csv')
employed_hh = raw_survey[raw_survey['employment_status'] > 0]
```

Automation and Abstraction 🤖

The Core Idea

Where possible, **automation** (avoiding human interaction) and **abstraction** (writing code flexible for many circumstances) should be embraced:

- Consider calculating $E(Y) = \sum_{x=x_{\min}}^{x_{\max}} E(Y|X = x) \cdot P(X = x)$
- One could tabulate X and compute expected values by hand
- Better: iterate through all values of X automatically (**AUTOMATION**)
- Best: write a function to do this for any two variables (**ABSTRACTION**)

```python
def conditional_expectation(df, outcome, conditioning_var):
    """Compute E[outcome] via law of iterated expectations."""
    probs = df[conditioning_var].value_counts(normalize=True)
    cond_means = df.groupby(conditioning_var)[outcome].mean()
    return (cond_means * probs).sum()
```

Why Functions?

Writing **functions** is central to both automation and abstraction:

- Separates and re-uses key code blocks
- Allows focus on perfecting code once, then reusing optimised versions
- Everything in a research project should be **replicable by machine**

> ⚠ **Warning**
>
> If you ever find yourself typing commands into a Python shell thinking “I’ll remember what I did”… you are in trouble. Write it down as a script!

LLMs and Workflow: Opportunities and Risks

AI coding assistants (GitHub Copilot, Claude, ChatGPT) have become a real part of modern workflows:

Opportunities:

- Accelerate boilerplate code writing
- Debug error messages quickly
- Translate across languages (e.g., Stata → Python)

Risks:

- Generated code may be subtly wrong or out of date
- Can obscure understanding if used as a crutch
- Attribution and reproducibility concerns

> **Important**
>
> **LLMs do not replace the need to understand your code.** Always verify, test, and understand what is generated.

Testing, Debugging & Profiling 🔍

Three Questions Every Programmer Faces

As soon as you have written some code:

Question	Term
My code runs, but does it do what it should?	**Testing**
My code does not run. Why?	**Debugging**
My code runs very slowly. Why?	**Profiling**

These are distinct activities requiring distinct approaches.

Testing

Testing is a **constant process** requiring vigilance:

- In empirical applications, test whether code returns correct values on **simulated data** where the answer is known
- Test **modularly**: each script should be tested standalone
- Similarly, each argument to a function should be tested
- Consider **proactively building in assertions**:

```
def compute_unemployment_rate(n_unemployed, n_labour_force):
    rate = 100 * n_unemployed / n_labour_force
    assert 0 ⩽ rate ⩽ 100, f"Rate {rate:.1f}% is out of bounds — check inputs"
    return rate
```

Modern Python: use `pytest` for systematic test suites.

Debugging

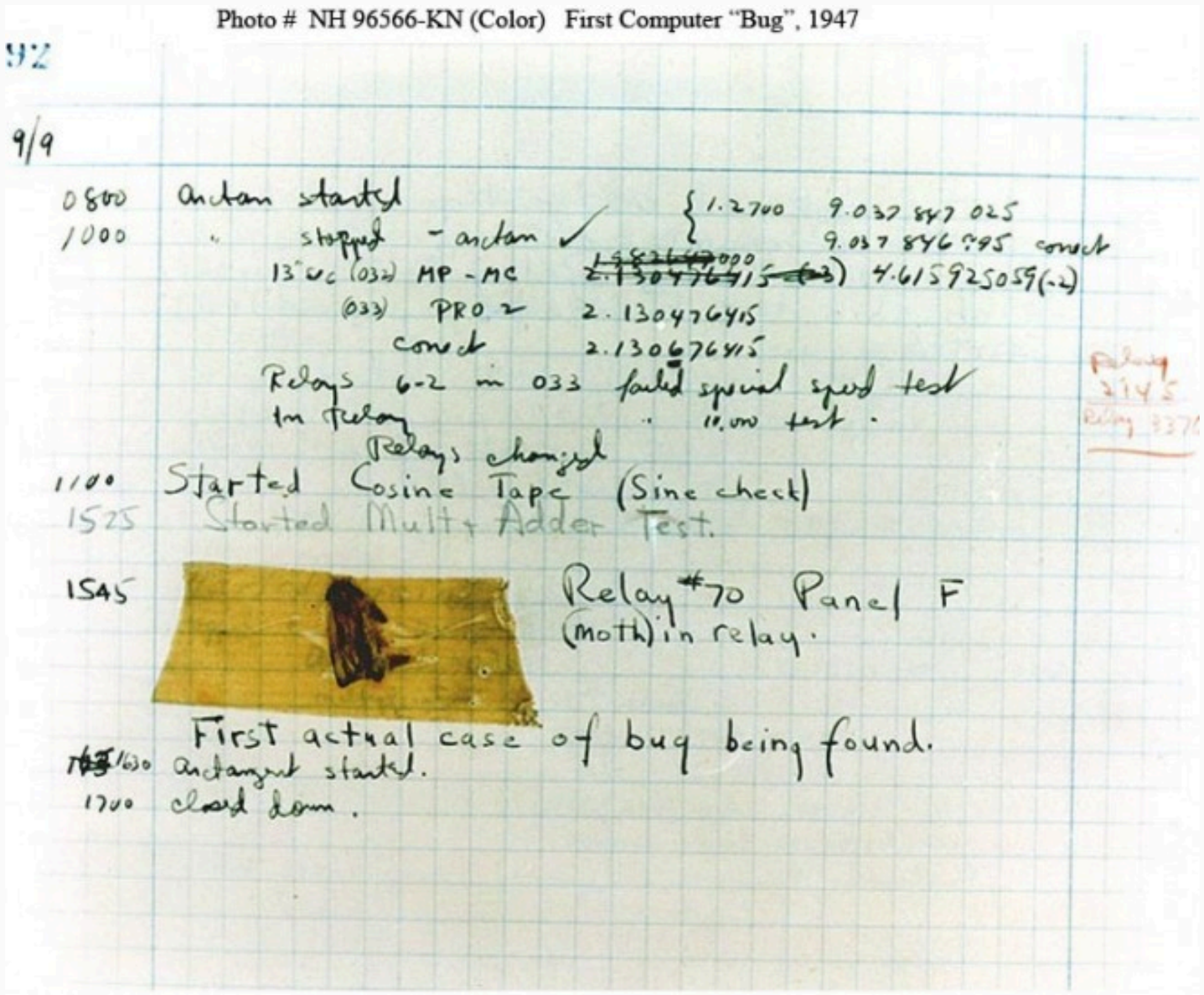
Debugging refers to finding errors that **prevent a script from running**:

- Read error messages carefully: they usually tell you what went wrong and where
- Consult the documentation as a first step
- As a last resort, paste the exact error into a search engine (or ask an LLM)
- Warnings: even when a script runs — should be investigated; they often signal inefficiencies or incorrect results

Tip

Do not think that just because a script runs, it is correct. Always return to testing.

The Famous Computer Bug



The famous "computer bug" from Grace Hopper's notebook (1947)

The term "bug" has been with us for almost 80 years.

Debugging in Practice: The Stack Trace

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In [4], line 3
      1 df = pd.read_csv('data/clean.csv')
      2 # Filter to treated observations
----> 3 treated = df[df['treatment_indicator'] == 1]

KeyError: 'treatment_indicator'
```

Read bottom to top: the error is `KeyError: 'treatment_indicator'`. The column does not exist. Check your column names with `df.columns`.

Wisdom of the Ancients

When you find that Stack Overflow has no answer...

Profiling

Profiling is about finding **hot spots**, ie bottlenecks, in code:

- Only begin optimising **after** you are sure the code is correct
- Many languages have profilers: Python has `cProfile` and `line_profiler`
- In Jupyter notebooks, use `%%time` or `%%timeit` to measure cell execution time
- **Do not over-optimise**: there is a real trade-off between machine time and programmer time

```
# Quick timing in Jupyter
%%timeit
result = [expensive_function(x) for x in large_list]

# Better: vectorised with numpy is usually faster
%%timeit
result = np.vectorize(expensive_function)(large_array)
```

An application to project management — Causal Forests 🌲

Today 📋

1. BRIEF Motivation: why heterogeneous treatment effects?
2. The ATE and CATE
3. Causal forests: a rough overview
4. Applying the workflow: project structure in practice

An Application: Causal Forests

To see an example of a complete stylised project, let's consider heterogeneity in some treatment effect:

- This is more on the **modelling side** of data science
- We will use a tool from causal machine learning known as **causal forests** (Athey, Tibshirani & Wager, 2019; Wager & Athey, 2018)
- We will briefly discuss the econometrics at a **high level**
- Note: this course is not about formal modelling – if you find this interesting, please consider taking further ML and causal econometrics courses (very happy to talk about this with you in office hours)!

The Starting Point: Average Treatment Effect (ATE) 📐

We are often interested in pinning down an **average treatment effect (ATE)**:

- Provides the impact of some treatment on outcomes in a population
- Formally defined as: $\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)]$
- If treatment is assigned exogenously (e.g., randomly): $\text{ATE} = \mathbb{E}[Y_i|T_i = 1] - \mathbb{E}[Y_i|T_i = 0]$
- *i.e.,* the treatment effect is simply the difference in mean outcomes between those treated and not
- This is often already a challenging parameter to nail down!

Conditional Average Treatment Effect (CATE) 🎯

But, we also care about **heterogeneity**: who gains and who loses from an intervention?

$$\text{CATE}(X) = \mathbb{E}[Y(1) - Y(0) \mid X]$$

- Measures the average treatment effect **conditioned on covariates** X
- Provides insight into how treatment effects vary across subgroups
- Policy relevance: targeting interventions to those who benefit most

> **Note**
>
> This is a significantly harder parameter to estimate than the ATE: standard regression struggles because of overfitting and the “curse of dimensionality”.

Why Can’t We Just Use OLS?

Consider estimating $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$ by running separate regressions within subgroups:

- With many covariates, cells become very thin
- Partitioning is arbitrary (which variables? how many bins?)
- OLS with interactions: degrees of freedom collapse rapidly

Machine learning offers tools to learn heterogeneous effects from data in a disciplined way — this is where causal forests come in.

Causal Forests: A Rough Overview 🌳

A **random forest** learns heterogeneity via **recursive partitioning** of the data:

- **Causal forests** extend random forests for estimating heterogeneous treatment effects (Wager & Athey, 2018; Athey, Tibshirani & Wager, 2019)
- We ask the data: which units are most similar in terms of their **treatment effects**?
- Effectively: split data to find units with the most similar effects, building up a “tree”
- Each decision tree is trained to partition data based on **covariates** and **treatment assignments**
- Each tree “learns” the relationship between features, treatment, and outcome

The Honest Tree

A key insight in causal forests: **honesty**

- Standard trees are “greedy”: they can overfit to the training data
- Causal forests use **sample splitting**: one half finds the best partition (tree structure), the other half estimates treatment effects within leaves
- This avoids in-sample bias and yields valid confidence intervals
- The forest then aggregates many such honest trees to reduce variance

> **Tip**
>
> This is what separates *causal* machine learning from predictive ML: we want valid inference, not just low prediction error.

Let’s get into this now…

We will explore real data from a large experiment (Oreopoulos, 2011).

Key References 📖

- **Wager, S. & Athey, S.** (2018). Estimation and Inference of Heterogeneous Treatment Effects using Random Forests. *Journal of the American Statistical Association*, 113(523), 1228–1242.
- **Athey, S., Tibshirani, J. & Wager, S.** (2019). Generalized Random Forests. *Annals of Statistics*, 47(2), 1148–1178.
- **Chernozhukov, V. et al.** (2018). Double/Debiased Machine Learning for Treatment and Structural Parameters. *The Econometrics Journal*, 21(1), C1–C68.
- **EconML documentation**: econml.azurewebsites.net
- **Oreopoulos, P.** (2011). Why Do Skilled Immigrants Struggle in the Labor Market? A Field Experiment with Thirteen Thousand Resumes. *American Economic Journal: Economic Policy*, 3(4):148–71.

Webscraping 

Today 📋

1. What is webscraping and why is it useful?
2. Considerations before scraping
3. Core tools: `requests`, `BeautifulSoup`, regex
4. More advanced tools: `Selenium`, `Playwright`
5. A worked example

What is Webscraping?

Essentially: **harvesting data stored on the web in irregular or highly dispersed format**

- When undertaking economic/data science analysis, we want regular data in rows and columns
- The web contains vast amounts of data that has never been packaged as a clean CSV
- Webscraping is the process of collecting and structuring it

> **Note**
>
> Two general steps: 1. **Loop** through nested URLs to reach many source HTML pages 2. **Parse** HTML (or JSON, XML) and format into a usable structure

Why is This Useful for Economists?

Often data is not stored as a CSV or standardised format:

- In some cases, data does not exist in **any centralised form**
- This opens up entirely new types of data we might not have considered
- The majority of economics papers now use **“novel” data** (i.e., not purely survey-based)

Real examples from my own work (simple real-use cases):

- Downloading, unzipping and merging 1,000+ DHS surveys, up-to-date at the moment of scraping
- Downloading all 30,000+ NBER working papers for text analysis
- Downloading election results: India, Philippines
- Collecting standardised statistics across many geographies and time periods

Economics Papers Using Scraped Data

Notable examples in the literature:

- **“The Billion Prices Project”** — Cavallo & Rigobon (2016): online prices scraped from hundreds of retailers to measure inflation in real time
- **“Nowcasting the Local Economy”** — Glaeser et al. (2017): Yelp data scraped to measure local economic activity
- **“Measuring Economic Policy Uncertainty”** — Baker, Bloom & Davis (2016): automated text analysis of newspaper archives
- **“Do Expiring Budgets Lead to Wasteful Spending?”** — Liebman & Mahoney (2017): federal contracting data scraped from USASpending.gov
- Many others: see Table 1 of Edelman (*JEP*, 2012)

Three Considerations Before You Scrape

Before starting any webscraping project, ask:

1. **Is the target standardised enough?**
2. **Are there legal limits?**
3. **Are there technical limits?**

Also: **do you even need to scrape?**

- Some websites provide data in a central download option already
- Many sites have **APIs** (sometimes free, sometimes paid) that make the task much easier
- The site IDEAS/RePEc is a great example of this
- Owners of sites are sometimes happy to provide data upon request

1. Is the Target Standardised Enough?

Webscraping requires interacting with **source code** (HTML or XML) of websites:

- **Much harder** if you are scraping many disparate websites rather than many pages within a single site
- **More challenging** if the source format changes over time (e.g., PDFs in early years, machine-readable in later years)
- Some websites are stored on **unstable servers** — plan for failures

> **Tip**
>
> A good webscraping script handles errors gracefully — use `try/except` blocks and build in retry logic. Never assume a request will succeed.

2. Are There Legal Limits?

Some websites **do not allow** webscraping — either because they consider information proprietary, or to prevent robot traffic from overloading their servers:

- The law is **complex and jurisdiction-dependent**
- There are legal precedents — e.g., *hiQ Labs v. LinkedIn* (US Ninth Circuit, 2022) found scraping publicly accessible data to be lawful, but this varies by context
- Where possible: **seek permission**, read **Terms of Service**
- Always check the `robots.txt` file (e.g., `https://example.com/robots.txt`)

```
# robots.txt example
User-agent: *
Disallow: /private/
Crawl-delay: 10
```

3. Are There Technical Limits?

Many websites have safeguards against excessive crawling (CAPTCHAs, rate limiting, IP blocking):

- Simple limit: **rate limiting** – slow down your requests
- IP blocking: more serious; could use proxies, but this is technically non-trivial
- **CAPTCHAs**_ "Completely Automated Public Turing test to tell Computers and Humans Apart" – designed specifically to block automated access

Important

Be a **polite scraper**: add delays between requests (`time.sleep()`), identify your bot in headers, and never hammer a server.

Core Webscraping Tools 🛠️

The Python Webscraping Stack

The typical workflow:

1. Use `requests` to download the HTML of a page
2. Use `BeautifulSoup` to parse and navigate the HTML structure
3. Use `re` (regex) for fine-grained text matching
4. Store results in a `pandas` DataFrame

HTML: A Quick Primer

To scrape a page, you need to understand its structure:

```
<html>
  <body>
    <div class="article">
      <h1>Article Title</h1>
      <p class="date">18 March 2026</p>
      <table id="results">
        <tr>
          <td>GDP Growth</td>
          <td>2.3%</td>
        </tr>
      </table>
    </div>
  </body>
</html>
```

Tags (`<div>`, `<p>`, `<table>`) have **attributes** (`class`, `id`) that help us locate data. Always use your browser's **developer tools** (right-click: Inspect) to explore the structure.

An Aside: Regular Expressions 🔤

Regular expressions (regex) are a standardised way to **match patterns in text**:

- We have seen and discussed these preeviously. Python has it's own implementation
- Most languages with string capabilities have regex libraries
- While precise syntax varies by language, there is a standard set of tools
- They can be simple but can become very complex

```
import re

text = "GDP grew by 2.3% in Q3 2025, compared to 1.8% in Q2 2025."

# Find all percentages
percentages = re.findall(r'\d+\.\d+%', text)
print(percentages)  # ['2.3%', '1.8%']

# Find years
years = re.findall(r'\b20\d{2}\b', text)
print(years)  # ['2025', '2025']
```

A useful resource for Python regex: docs.python.org/3/howto/regex.html

Let’s see an example…

We will implement a webscrape in real time – this can be quite simple in some cases!

`BeautifulSoup`: Parsing HTML

A basic (fictitous) example…

```
from bs4 import BeautifulSoup

soup = BeautifulSoup(html_content, 'html.parser')

# Find a single element
title = soup.find('h1').text

# Find all elements matching a selector
all_rows = soup.find_all('tr')

# CSS selectors (often cleaner)
date = soup.select_one('p.date').text

# Navigate the tree
article_div = soup.find('div', class_='article')
table = article_div.find('table', id='results')

# Extract text from all cells in a table
cells = [td.text.strip() for td in table.find_all('td')]
```

Let’s go back to our example and see that this can help!

Beyond Static Pages: JavaScript-Rendered Content

Many modern websites **render content dynamically using JavaScript**_ the HTML you download with `requests` may be empty or incomplete:

- Single-page applications (React, Vue, Angular)
- Lazy-loaded content (data loaded as you scroll)
- Content behind login forms

Solution: "headless browsers": automate a real browser that executes JavaScript:

- **Selenium**: the classic, widely used
- **Playwright**: modern, faster, more reliable; increasingly preferred (full transparency, I have not used this one…)

Selenium / Playwright: When to Use Them

```python
from playwright.sync_api import sync_playwright

with sync_playwright() as p:
    browser = p.chromium.launch(headless=True)
    page = browser.new_page()

    page.goto("https://dynamic-site.com/data")

    # Wait for specific content to load
    page.wait_for_selector('table#results')

    # Get the fully rendered HTML
    html = page.content()

    browser.close()

# Now parse with BeautifulSoup as normal
soup = BeautifulSoup(html, 'html.parser')
```

A Note on LLMs and Webscraping (2025)

A new approach is gaining traction: using **LLMs to extract structured data** from HTML directly:

- Tools like Crawl4AI, Firecrawl, and the Anthropic API can be used to parse unstructured HTML into structured data
- Particularly useful when website structures vary or change frequently
- Still requires careful prompting and validation

```
# Conceptual example: LLM-based extraction
import anthropic

client = anthropic.Anthropic()
response = client.messages.create(
    model="claude-sonnet-4-20250514",
    messages=[{
        "role": "user",
        "content": f"Extract all paper titles and authors from this HTML as JSON:\n\n{html_snippet}"
    }]
)
```

Note

This can be powerful but is slower, more expensive, and requires validation: **use with care**.

Where to From Here? 🚀

This unit has discussed a range of tools broadly within a programming workflow. But as always, mastery comes from practice:

- **Final Project**: A nice place to practice thinking about workflow in a formal way. Advice: **do what is interesting to you**
- **Further reading**:
  - Coding for Economists — Turrell (2023)
  - EconML documentation
  - Beautiful Soup documentation
  - Playwright for Python
- **Questions?** Write me at dcc213@exeter.ac.uk or during office hours (I will maintain the Monday block for a few weeks, though feel free to write for other times)

Fin!

Thank you!